

What Is the Turing Test?

In the Turing Test, a computer and human are asked questions to determine which is human. The computer passes if it is indistinguishable from the human. Here's how the test works, a brief history, variations, limitations and how it's used today.



Written by [Eric Kleppen](#)



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UPDATED BY
[Matthew Urwin](#) | Jan 15, 2025

The Turing Test, one of the most discussed methods for assessing [artificial intelligence](#) (AI), dates back to the 1950s. It grew out of a thought experiment devised by computer scientist Alan Turing in which he devised what he initially named The Imitation Game. This test pits human respondents against a machine to test the machine's ability to exhibit human-like responses and intelligence. To this day, the Turing Test is widely considered a benchmark for measuring the success of [AI research](#).

What Is the Turing Test?

The Turing Test is a method that tests a machine's ability to exhibit human-like responses and intelligence. It has popularly been used as a benchmark testing method to assess the development of artificial intelligence (AI) systems.

The Turing test: Can a computer pass for a human? - Alex Gendler



The Turing Test: Can a Computer Pass for a Human? — Alex Gendler | Video: TED-Ed

How Does the Turing Test Work?

The Turing Test is performed by placing a human in one room and a machine in another. Then a judge, or panel of judges, addresses each room with questions regarding any topic to which a human should be able to respond. If the machine passes Turing's test, it shows the machine's ability to process human syntax and semantics, which is thought to be a step toward creating [artificial general intelligence](#).

Regardless of a computer's ability to pass the Turing Test, there is no real way for us to tell whether or not a machine truly understands [human semantics](#). The test simply judges machines on their ability to converse with human-like eloquence, not human-like understanding. This limitation has led some [AI researchers](#) to argue the Turing Test is less relevant than it used to be.

Has Anything Ever Passed the Turing Test?

While no machine has ever passed the Turing Test flawlessly, several machines have fooled judges to some extent. In 1966, MIT professor Joseph Weizenbaum created a machine named ELIZA, which is regarded as one of the first computers to have fooled a judge. The chatbots Eugene Goostman and GPT-4 also passed the Turing Test in 2014 and 2024, respectively.

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Turing Test Example Questions

While there is no official list of Turing Test questions, a judge would likely ask questions that relate to human experiences like emotions and maturation, or [linguistic riddles](#) that could be difficult for a machine to parse. Here are some questions to ask if you find yourself judging a Turing Test:

- What is your most memorable childhood event and how has that impacted you today?
- Describe yourself using only colors and shapes.
- Describe why time flies like an arrow but fruit flies like a banana?
- How do you feel when you think about your upbringing and what makes you feel that way?
- What historical event changed you the most and where were you when it happened?
- Which of the previous questions was the most difficult to answer and why?

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History of the Turing Test

Alan Turing is considered one of the pioneers of computer science and artificial intelligence. His original proposal for the Turing Test was his 1950 paper “[Computing Machinery and Intelligence](#).” The paper focuses on the question, “Can machines think?” To answer this question, Turing proposed a test in which a human judge would engage in a [natural language conversation](#) with both a human and a machine, without knowing which was which. If the judge couldn’t distinguish the machine from the human, the machine passed Turing’s test.

Over the following decades, the field of AI has made [significant progress](#) and the Turing Test evolved. The Loebner Prize Turing Test began in 1990 and is recognized as one of the most prominent versions of the Turing Test. In 2010, a computer dubbed Bruce Wilcox successfully fooled one judge a single time for the Loebner Prize. Since then, other machines have fooled judges and won the Loebner Prize.

The Loebner Prize stopped being awarded in 2020, but this hasn’t discouraged further attempts to pass the Turing Test. In 2024, [researchers claimed](#) that [GPT-4](#) passed the Turing Test since it tricked participants into thinking it was human 54 percent of the time. For reference, the [ELIZA chatbot](#) fooled participants only 22 percent of the time. The last AI system to pass the Turing Test before GPT-4 was [Eugene Goostman](#) — a [chatbot](#) that tricked 33 percent of judges into thinking it was a 13-year-old Ukrainian.

Turing Test: Variations and Alternatives

Since its inception, Turing’s test has undergone slight changes but the goal has always remained the same — to [evaluate artificial intelligence](#). Although Turing himself never specified the amount of time given to the judge, more recent versions of the test, like the Loebner Prize Turing Test, rule that a machine has passed if the judge cannot determine which room has a human and which room has a machine after a question-and-answer period of 25 minutes.

As artificial intelligence technology improved, others have devised variations on the Turing Test. All of these tests have shortcomings and none are as well-known as the Turing Test. That said, some of the more interesting variations include the Reverse Turing Test, the Marcus Test and the Lovelace Test 2.0.

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The Reverse Turing Test

In the reverse Turing Test, the subjects attempt to [appear as a computer](#) rather than a human. The goal is to trick a computer into believing it’s not interacting with a human. CAPTCHA security measures that you’ve likely encountered when signing onto a website is a form of the reverse Turing Test, which means the machine is trying to evaluate if it’s interacting with an actual human or another machine.

The Marcus Test

In the Marcus Test, devised by cognitive scientist Gary Marcus, subjects watch TV shows or YouTube videos and respond to questions about the content. For a machine to understand an ongoing television program, the machine must comprehend the events over time. This evaluates an AI’s [human-like understanding](#).

The Lovelace Test 2.0

Lastly, the Lovelace Test 2.0, named after mathematician Ada Lovelace, looks for [computational creativity](#). This test has recently become more relevant due to the advancements of text-to-image technology like [MidJourney](#) and [OpenAI’s DALL·E2](#). In the Lovelace Test, the judge comes up with a set of constraints that they expect the machine to be unable to meet. If the judge cannot tell which creation is from a machine, they may come up with a more difficult set of constraints in the next round of testing.

What Are the Limitations of the Turing Test?

Although Alan Turing came up with an influential test while considering whether or not machines can think, Turing's test is not a sufficient indicator of artificial intelligence. Not only does Turing's test fail to account for whether or not a machine understands its input and output, it also accounts for neither a machine's ability to [recognize patterns](#) nor its ability to apply common knowledge or sense.

Beyond the limitations of the test itself, many AI researchers feel the Turing Test is irrelevant today. With advances in [data science](#) and [cloud computing](#), there's been a growing focus on [natural language processing](#) (NLP) and creating large language models like [ChatGPT](#), [BERT](#) and now [Gemini](#) (formerly [Bard](#)). Over the past decade, NLP technology has improved dramatically, thereby allowing machines to better understand and generate human-like language with increasing accuracy. In 2021, Google created a chatbot called [LaMDA](#) that was so good, one of the AI researchers working on it believed it [achieved sentience](#).

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How Is the Turing Test Used Today?

Whether or not the Turing Test is truly relevant remains a hotly debated topic for AI researchers. The past decade has seen [significant advances](#) in the field of AI. These advances have been made possible by the development of more sophisticated AI [algorithms](#), access to more powerful computing [hardware](#), as well as a focus on [natural language processing](#) and [multimodal capabilities](#). As a result, machines are becoming increasingly capable of exhibiting intelligent behavior indistinguishable from humans.

The rapid rise of [generative AI](#) has led to technologies that can produce realistic text responses, images, music and other content. In the creative sectors, [AI art](#) has exposed how the [Turing Test](#) fails to discern between [human- and AI-generated art](#). These developments have encouraged broader calls for an [updated Turing Test](#).

That said, many feel AI is still a long way from achieving human-like general intelligence and the Turing Test remains one of the many ways in which humans can evaluate a dimension of an [AI's abilities](#). For example, the Turing Test's capacity to [measure "indistinguishability"](#) makes it useful for applications like gauging the effectiveness of [facial recognition](#) technology and ensuring the safety of [self-driving cars](#).

And when companies like Google create [large language models](#) and push the boundaries of [chatbot technology](#), they still use human evaluators to ask a series of questions to determine its abilities. In this way, some form of Alan Turing's thought experiment remains culturally relevant to the [advancement of artificial intelligence](#).

Frequently Asked Questions

Has anything passed the Turing Test?



There has never been a machine that has perfectly passed the Turing Test. However, there have been some AI machines that have been argued to have passed the Turing Test or have fooled testing judges, including the chatbots ELIZA, Eugene Goostman and ChatGPT.

What questions are asked in the Turing Test?



Questions asked in the Turing Test involve those able to be answered by a human, such as questions about human experiences and emotion or linguistic riddles.

There is no official set of questions used in the Turing Test, but some example questions can include:

- What is your most memorable childhood event and how has that impacted you today?
- Describe why time flies like an arrow but fruit flies like a banana?
- What historical event changed you the most and where were you when it happened?

Is the Turing Test still used today?



Variations of the Turing Test are still used today to assess artificial intelligence research, including the reverse Turing Test and the Marcus Test. Turing Tests are useful for determining whether AI can perform tasks just as well as — if not better than — humans, such as recognizing faces or fulfilling customer service roles.

Has any human failed the Turing Test?



Yes, humans have been known to [make the mistake](#) of identifying their human chat partners as machines. In addition, the ELIZA, Eugene Goostman and ChatGPT chatbots all successfully fooled participants into thinking they were human at least 22 percent of the time.

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